SPRiF: Spectral Memory and Scalar Reset for Spiking Neurons

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***Abstract*—A network may need an early cue long after the neurons that received it have fired. In a leaky integrate-and-fire neuron, the cue and each subsequent reset contribute to the same voltage trace. We introduce SPRiF, a five-state neuron that stores input current in a spectral memory and resets the membrane that reads it. One real decay and a damped rotation provide the temporal basis. Two learned projections drive the membrane, one immediately and one through an auxiliary filter delayed by one step. We derive the membrane drive in terms of the input history and identify the additional spike-history term produced by a reset along the memory input direction. We evaluate SPRiF on five temporal benchmarks and test reset placement and mode organization with matched controls on two of them. Preserving memory at reset gave mean accuracy gains of 3.37 percentage points on permuted sequential MNIST and 4.56 on Spiking Heidelberg Digits over the memory-reset control. On delayed match-to-sample, SPRiF retained 99.42% mean accuracy under 40 forced-spike timesteps after clean training. Training with up to eight intervention timesteps yielded 99.33% under the same stress, averaged over five delays including an unseen 2500-ms delay.**

***Index Terms*—Spiking neural networks, temporal processing, spectral memory, neuron dynamics, scalar reset.**

I. Introduction

In a delayed decision, the first cue may be gone by the time the second arrives. Spikes emitted between the cues serve two roles in a recurrent spiking neural network (SNN). They reset the membranes that fired and enter the recurrent current on the next step. In the leaky integrate-and-fire (LIF) model, an input pulse and a subsequent reset leave traces in the same membrane voltage, both governed by the same decay. The signal available at the second cue therefore depends on the intervening spikes as well as the first cue.

A neuron with several state variables can assign these roles differently. Adaptive neurons retain a history of activity [1], [2], heterogeneous neurons and synapses distribute temporal integration across time scales [3], [4], resonant dynamics produce lag-dependent responses [5]–[7], and coupled compartments distribute integration across states [8], [9]. PMSN separates linear hidden compartments from a firing compartment [10], while DMP-SNN maintains a memory shared across a layer [11]. Reset itself can also be softened, adapted or removed [12]–[14]. These developments motivate a joint question about the organization of a fixed number of states within one neuron: which temporal responses can the membrane read, and how does a spike change those responses?

SPRiF assigns three coordinates to a spectral memory, one to a filter and one to the membrane. A pulse excites a decaying trace and a damped rotation. The membrane reads a projection of the current spectral state and a filtered projection from the preceding step. These two paths allow the neuron to change the weight of older inputs while holding the instantaneous response fixed. The local reset acts on the membrane alone. For a specified current sequence, we derive the kernel that maps input to membrane drive and identify the extra spike-history term introduced when reset also acts on the spectral state.

We evaluate SPRiF on five temporal benchmarks and test the two design choices on permuted sequential MNIST (PS-MNIST) and Spiking Heidelberg Digits (SHD). Two controls retain the same five states, eleven learned local coefficients and layer widths. One adds a reset to spectral memory. The other replaces the spectral block with three driven real decays. A second experiment, Spike-Intervention Delayed Match-to-Sample (SI-DMS), forces spikes between two cues. It tests whether the full recurrent network can still compare the cues when its spike history changes.

II. Related Work

A. States That Carry Temporal Information

Adaptive neurons such as ASRNN and adLIF retain activity in a variable that influences later firing [1], [2]. DA-LIF introduces independently learnable decays for spatial and temporal tuning [15]. Temporal diversity can also be distributed across neurons: heterogeneous neuronal parameters support different integration times [3], and bimodal time-scale networks explicitly combine populations with short and long time constants [16]. HetSyn instead assigns time constants to individual synapses, placing temporal integration in synaptic currents [4]. Conductance-based models obtain multiple time scales through fast, slow and ultra-slow conductances [17]. These approaches broaden the temporal dynamics available to a network; SPRiF places a decaying mode and a rotational pair together within each neuron.

In resonant neurons, the phase of a coupled state affects the response to a past input [5], [6]. PRF combines resonant membrane dynamics with a decoupled reset construction for parallel sequence processing [18]. Compartmental models offer another organization: Dendrify provides reduced models with dendritic integration and local spikes [19], TC-LIF couples dendritic and somatic states [8], and dendritic heterogeneity gives different branches different time constants [9]. A close relative of SPRiF is D-RF, whose resonant dendrites feed a somatic readout and whose threshold depends on spike history [7]. Its dendrites also evolve without subtractive output reset. SPRiF gives the membrane two learned projections of its spectral states, with the second acting through an auxiliary filter one step later.

B. Spectral Memory and Output Reset

State-space models motivate both the parameterization of spiking dynamics and their organization within a sequence

model. SiLIF transfers learnable discretization and logarithmic parameterization to spiking neurons, with a complex-state variant that supports oscillatory regimes [20]. SpikingSSMs combine state-space sequence processing with sparse spiking outputs [21]. SPRiF specifies a real pole and a complex-conjugate pair directly, then learns how their states reach the membrane through immediate and delayed projections.

PMSN separates the dynamics of its hidden compartments from discharge at its output [10]. Unrolling its diagonalized hidden transition gives a convolution from injected current to the drive at the output compartment. Its parallel output update requires a separate construction, temporal-decoupled reset, which operates on rectified current and discharges integer multiples of the threshold. SPRiF uses a leaky membrane with signed drive and one threshold subtraction per binary spike. This choice gives a closed form for the drive to the membrane while retaining a sequential firing decision. DMP-SNN makes a different architectural choice by sharing memory within a layer [11]. Our matched controls examine the temporal modes and reset placement within one neuron.

C. Spike History in Computation and Training

Reset design determines how firing changes the state available at later timesteps. Guo *et al.* use a soft reset and membrane-potential rectification to reduce information loss [12]; AR-LIF adapts reset through the relation between input and output [13]. RPLIF models refractory effects through spike-triggered threshold dynamics [22], while Zhang analyzes reset and refractory periods from the perspective of binary-activated recurrent networks [23]. PSN removes reset to obtain parallelizable temporal computations [14]. SPRiF retains a subtractive membrane reset and analyzes which additional history term appears when that reset is extended to spectral memory.

NiSNN-A writes LIF voltage as the difference between accumulated input and accumulated reset [24]. Its NiLIF update estimates the unknown spike train from the input-only accumulation to obtain a noniterative computation. In SPRiF, spikes are generated by thresholding the actual membrane candidate at each step. The local response to a realized spike can then be derived from the state equations.

IML-Spikeformer uses input-dependent thresholds for multilevel firing and examines the resulting firing-rate distributions [25]. STDL changes which dependencies are retained during learning to fit a training-memory budget [26]. In the comparisons below we keep surrogate-gradient backpropagation through time (BPTT) and change the forward dynamics of the neuron.

III. SPRiF Neuron: Model and Analysis

A. From a Reset-Coupled Trace to Separate State Roles

To fix the timing convention, let v_t be a LIF membrane immediately after the current reset. Its candidate, spike, and stored voltage are

$$\widetilde{v}_t = \beta v_{t-1} + I_t, \qquad z_t = H(\widetilde{v}_t - \theta), \qquad v_t = \widetilde{v}_t - \theta z_t, \tag{1}$$

where $0 < \beta < 1$, $\theta > 0$, and $H(a) = 1$ for $a \geq 0$ and zero otherwise. Unrolling gives

$$v_t = \beta^{t+1} v_{-1} + \sum_{j=0}^{t} \beta^{t-j} I_j - \theta \sum_{j=0}^{t} \beta^{t-j} z_j. \tag{2}$$

A unit of current at step j contributes β^{t-j} at step t. A spike at the same step contributes $-\theta\beta^{t-j}$. The two contributions differ in sign and scale but have the same time course. SPRiF retains membrane reset and gives the membrane access to a second history, a filtered record of the input current whose coordinates are preserved at discharge.

B. Spectral Memory and Membrane Update

For an input vector $\mathbf{s}_t$ and a recurrent spiking layer, the current vector is

$$\mathbf{I}_t = W_{\text{in}} \mathbf{s}_t + W_{\text{rec}} \mathbf{z}_{t-1} + \mathbf{b}. \tag{3}$$

The recurrent term is omitted for a feedforward layer. The equations below concern one neuron and suppress its index. Its current I_t is the corresponding component of $\mathbf{I}_t$, and it stores $\mathbf{x}_t \in \mathbb{R}^3$, $h_t \in \mathbb{R}$, and $u_t^0 \in \mathbb{R}$:

$$\begin{cases} \mathbf{x}_t = A\mathbf{x}_{t-1} + BI_t, \\ h_t = \eta_1 h_{t-1} + \bar{\mathbf{g}}_1^\top \mathbf{x}_t, \\ \widetilde{u}_t^0 = \eta_0 u_{t-1}^0 + h_{t-1} + \mathbf{g}_0^\top \mathbf{x}_t, \\ z_t = H(\widetilde{u}_t^0 - \theta), \\ u_t^0 = \widetilde{u}_t^0 - \theta z_t. \end{cases} \tag{4}$$

The transition and input vector are

$$A = \begin{bmatrix} \alpha & 0 & 0 \\ 0 & \rho\cos\omega & -\rho\sin\omega \\ 0 & \rho\sin\omega & \rho\cos\omega \end{bmatrix}, \qquad B = \begin{bmatrix} 1-\alpha \\ 1-\rho \\ 0 \end{bmatrix}. \tag{5}$$

The two directly learned readouts form

$$\bar{G} = \begin{bmatrix} \mathbf{g}_0^\top \\ \bar{\mathbf{g}}_1^\top \end{bmatrix} \in \mathbb{R}^{2\times 3}. \tag{6}$$

For finite raw parameters, the constraints are enforced by

$$\begin{aligned} &\alpha = \sigma(\alpha_{\text{raw}}), \qquad \rho = \sigma(\rho_{\text{raw}}), \\ &\omega = \pi\sigma(\omega_{\text{raw}}), \quad \eta_i = \sigma(\eta_{i,\text{raw}}), \quad i \in \{0,1\}. \end{aligned} \tag{7}$$

Here σ is the logistic sigmoid. Thus $\alpha, \rho, \eta_i \in (0,1)$ and $\omega \in (0,\pi)$. The six entries of $\bar{G}$ are unconstrained real coefficients. The threshold is a fixed task setting.

Figure 1 contrasts the state organization of LIF and SPRiF. In SPRiF, two routes connect $\mathbf{x}_t$ to firing. The direct projection enters $\widetilde{u}_t^0$ immediately. The second projection updates h_t, which first enters the membrane at $t+1$. At the current step the membrane reads h_{t-1}. All intrinsic states start at zero for an independent sequence.

The filter-to-membrane coupling in Eq. (4) has unit gain. A separate constant gain κ would be redundant with the unconstrained input projection of the filter. For a filter $u_t^1 = \eta_1 u_{t-1}^1 + \mathbf{g}_1^\top \mathbf{x}_t$ read out as κu_{t-1}^1, set

$$h_t = \kappa u_t^1, \qquad \bar{\mathbf{g}}_1 = \kappa \mathbf{g}_1, \qquad h_{-1} = \kappa u_{-1}^1. \tag{8}$$

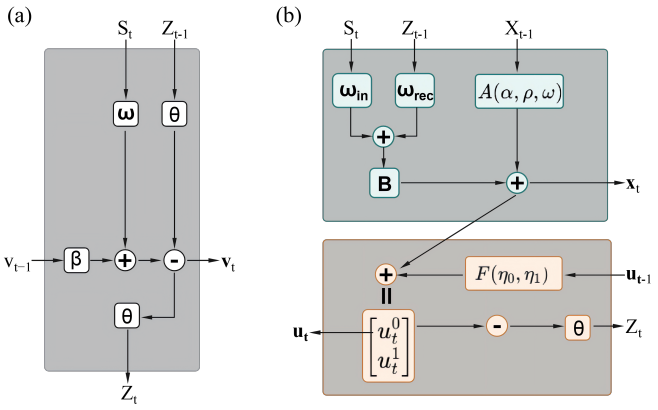


Fig. 1. State organization of LIF (left) and SPRiF (right). LIF integrates input and reset in the same membrane variable. SPRiF separates spectral memory, governed by $A(\alpha, \rho, \omega)$ and B, from the auxiliary filter and membrane dynamics. The spectral state supplies the drive for firing, while the spike-triggered reset acts on the membrane.

Substitution gives the same forward membrane voltage and spikes for fixed coefficients, including at $\kappa = 0$. SPRiF learns $\bar{\mathbf{g}}_1$ directly. This leaves five decay or frequency parameters and six projection coefficients per neuron.

C. How an Input Pulse Reaches the Membrane

The poles of the spectral transition are

$$\operatorname{spec}(A) = \{\alpha, \rho e^{\mathrm{i}\omega}, \rho e^{-\mathrm{i}\omega}\}. \tag{9}$$

For a unit impulse $I_0 = 1$, zero state at $t = -1$, and zero later current,

$$A^k B = \begin{bmatrix} (1-\alpha)\alpha^k \\ (1-\rho)\rho^k \cos(k\omega) \\ (1-\rho)\rho^k \sin(k\omega) \end{bmatrix}, \qquad k \geq 0. \tag{10}$$

The first coordinate retains one sign. In the other two, ρ sets the decay per step and ω sets the phase advance. For example, at $\omega = \pi/2$ the cosine coordinate equals $1-\rho$ at lag zero and $-(1-\rho)\rho^2$ two steps later. A projection onto that coordinate can therefore give the same pulse opposite weights according to its age.

The auxiliary filter accumulates a second projection of this trajectory. For the same pulse its response is

$$h_k = \sum_{\ell=0}^{k} \eta_1^{k-\ell} \bar{\mathbf{g}}_1^\top A^\ell B. \tag{11}$$

Write the membrane drive as

$$d_t = \mathbf{g}_0^\top \mathbf{x}_t + h_{t-1}. \tag{12}$$

Substituting the spectral and filter responses gives the kernel

$$\begin{aligned} k_d[k] = {} & \mathbf{g}_0^\top A^k B \\ & + \sum_{\ell=0}^{k-1} \eta_1^{k-1-\ell} \bar{\mathbf{g}}_1^\top A^\ell B, \end{aligned} \tag{13}$$

The sum is empty at $k = 0$. In particular, $k_d[0] = \mathbf{g}_0^\top B$ and $k_d[1] = \mathbf{g}_0^\top AB + \bar{\mathbf{g}}_1^\top B$. Adjusting the second projection can change the response one step after a pulse while leaving its onset unchanged. Writing ξ for a unit delay gives the transfer function

$$\mathcal{K}_d(\xi) = \left(\mathbf{g}_0^\top + \frac{\xi}{1-\eta_1\xi} \bar{\mathbf{g}}_1^\top \right) (I_3 - \xi A)^{-1} B. \tag{14}$$

For fixed coefficients and zero initial state, convolving k_d with the realized current sequence gives the exact drive. The membrane then integrates this drive and its own reset history. For any initial membrane voltage,

$$\begin{aligned} u_t^0 = {} & \eta_0^{t+1} u_{-1}^0 + \sum_{j=0}^{t} \eta_0^{t-j} d_j \\ & - \theta \sum_{j=0}^{t} \eta_0^{t-j} z_j. \end{aligned} \tag{15}$$

Equation (15) assigns the membrane its own leak η_0. The drive that precedes this integration has the poles of the spectral block and the auxiliary filter, subject to cancellation by the learned projections.

The complex pair contributes the quadratic factor $1 - 2\rho\cos(\omega)\xi + \rho^2\xi^2$ to the denominator of Eq. (14). If this factor survives in the readout, a block containing only positive real decays and the same auxiliary filter cannot reproduce the kernel at all lags. Its denominator has only real linear factors. The distinction is an intrinsic oscillatory mode, since signed sums of real decays can themselves change sign. Section IV-C defines a control with three driven real modes.

D. Reachability and Bounds on the Local States

Current enters two spectral coordinates directly (Eq. (5)). Rotation couples one of them to the third. The resulting three-dimensional state can be reached with a scalar current.

Proposition 1 (Driven spectral coordinates): For $\alpha, \rho \in (0, 1)$ and $\omega \in (0, \pi)$, the pair (A, B) in Eq. (5) is controllable.

Proof: Let $\mathcal{C} = [B, AB, A^2B]$. Expanding its determinant gives

$$\begin{aligned} \det \mathcal{C} = {} & (1-\alpha)(1-\rho)^2 \rho \sin\omega \\ & \times (\alpha^2 - 2\alpha\rho\cos\omega + \rho^2). \end{aligned} \tag{16}$$

Every factor is positive, since the last factor equals $|\alpha - \rho e^{\mathrm{i}\omega}|^2$. Thus $\mathcal{C}$ has rank three. ∎

From zero state, three appropriately chosen scalar inputs can therefore reach any point in the spectral state space. Reachability concerns the state before projection. A learned readout can suppress a reachable mode, and a nearly singular controllability matrix can make some directions difficult to excite.

For fixed parameters and a specified current sequence, define $\mathbf{m}_t = [\mathbf{x}_t^\top, h_t]^\top$. The memory pathway is

$$\mathbf{m}_t = F\mathbf{m}_{t-1} + LI_t, \tag{17}$$

with

$$F = \begin{bmatrix} A & \mathbf{0} \\ \bar{\mathbf{g}}_1^\top A & \eta_1 \end{bmatrix}, \qquad L = \begin{bmatrix} B \\ \bar{\mathbf{g}}_1^\top B \end{bmatrix}. \tag{18}$$

The block-triangular structure gives

$$\operatorname{spec}(F) = \{\alpha, \rho e^{\mathrm{i}\omega}, \rho e^{-\mathrm{i}\omega}, \eta_1\}. \tag{19}$$

Each pole is strictly inside the unit disk. The learned projection into h_t can amplify a transient even though every mode decays.

Proposition 2 (Bounded local states): Assume fixed finite readouts, finite initial states, $\theta > 0$, the ranges in Eq. (7), and $|I_t| \le M$ for all $t \ge 0$. Then $\mathbf{x}_t$, h_t, and u_t^0 in Eq. (4) remain bounded.

Proof: Let $q = \max(\alpha, \rho) < 1$. The rotational part of A^k is a rotation scaled by ρ^k, so $\|A^k\|_2 = q^k$. Summing the geometric series for the forced spectral recurrence gives a uniform bound X on $\|\mathbf{x}_t\|_2$. Applying the same argument to the filter gives a bound H_* on $|h_t|$, where

$$\begin{aligned} X &= \|\mathbf{x}_{-1}\|_2 + \frac{\|B\|_2 M}{1-q}, \\ H_* &= |h_{-1}| + \frac{\|\bar{\mathbf{g}}_1\|_2 X}{1-\eta_1}. \end{aligned} \tag{20}$$

It follows that $|d_t| \le \|\mathbf{g}_0\|_2 X + H_*$. Since each spike subtracts at most θ, the expansion in Eq. (15) yields

$$|u_t^0| \le |u_{-1}^0| + \frac{\|\mathbf{g}_0\|_2 X + H_* + \theta}{1-\eta_0}. \tag{21}$$

All three bounds are finite. ∎

For a prescribed current sequence, a perturbation within the memory and filter coordinates propagates as $\delta\mathbf{m}_{t+k} = F^k \delta\mathbf{m}_t$. Poles near the unit circle permit slow decay along this path. If a spike changes a subsequent current, an additional input-driven response must be included.

E. Where a Spike Enters the Memory Dynamics

Use superscripts $-$ and $+$ for values immediately before and after the current local reset. Equation (4) gives

$$\mathbf{x}_t^+ = \mathbf{x}_t^-, \qquad h_t^+ = h_t^-, \qquad u_t^{0,+} = \widetilde{u}_t^0 - \theta z_t. \tag{22}$$

A spike changes stored voltage immediately. All four memory and filter coordinates have zero reset increment, so for a given current sequence the drive d_t contains no direct contribution from past local resets.

To see what changes when a spike also acts on memory, consider the control in Eq. (29). After both projections have read $\mathbf{x}_j^-$, a spike at j adds $-\theta\widehat{B}$ to the stored spectral state, where $\widehat{B} = B/\|B\|_2$. An input pulse of size $-\theta/\|B\|_2$ would produce the same spectral displacement. Its timing within the step would differ, however, because an input pulse enters h_j before the reset occurs. Under a unit pulse, that filter update contributes $\eta_1^{k-1}\bar{\mathbf{g}}_1^\top B$ to the drive at lag $k \ge 1$. Removing this contribution from Eq. (13) and scaling by the signed pulse amplitude gives the kernel r from a spike to the membrane drive. The current drive has already been computed, so $r[0] = 0$. At later lags,

$$r[k] = -\frac{\theta}{\|B\|_2}\left(k_d[k] - \eta_1^{k-1}\bar{\mathbf{g}}_1^\top B\right), \quad k \ge 1. \tag{23}$$

For zero initial states, superposition in the memory and filter recurrences now gives

$$d_t = \sum_{k=0}^{t} k_d[k]\, I_{t-k} + \sum_{k=1}^{t} r[k]\, z_{t-k}. \tag{24}$$

The second sum is the direct contribution of memory reset in `MEMORY_RESET5`. In SPRiF its kernel is identically zero by Eq. (22). Each identity uses the model's own fixed coefficients and realized currents and spikes. The membrane in both models still receives its scalar reset.

A spike can also return through the network. Holding the external layer input fixed, a change in the spike vector at t changes the next recurrent current by

$$\delta\mathbf{I}_{t+1} = W_{\mathrm{rec}}\, \delta\mathbf{z}_t. \tag{25}$$

This change enters the first sum of Eq. (24) through the same kernel as any other current. It is distinct from the direct memory reset in the second sum. SI-DMS tests the joint effect of overwriting membrane candidates, firing, reset and recurrent propagation on a delayed decision.

F. Learning and Parameter Count

The forward pass uses a hard threshold. During BPTT, its derivative is replaced by the multi-Gaussian surrogate [1]

$$\begin{aligned} \psi(v) = \gamma\big[&a_0\mathcal{N}(v-\theta; 0, \sigma_0^2) \\ &- a_1\mathcal{N}(v-\theta; \mu_1, \sigma_1^2) \\ &- a_1\mathcal{N}(v-\theta; -\mu_1, \sigma_1^2)\big], \end{aligned} \tag{26}$$

where $\mathcal{N}(\cdot; \mu, \sigma^2)$ is the normalized Gaussian density, $\sigma_0 = 0.5$, $\mu_1 = 0.5$, $\sigma_1 = 3.0$, $a_0 = 1.15$, $a_1 = 0.15$, and $\gamma = 0.5$. Gradients pass through the recurrent spike and the membrane reset within each BPTT segment (Section IV-B).

Each neuron stores (x^0, x^1, x^2, h, u^0), for five state values. It learns the five raw variables associated with $\alpha, \rho, \omega, \eta_0, \eta_1$ and six entries of $\bar{G}$, for eleven dynamics coefficients. Thresholds are fixed, and synaptic weights, biases, and output parameters are separate.

For a recurrent layer with input width m and hidden width n, a dense input matrix, dense recurrent matrix, and one current bias per neuron contribute

$$N_{\mathrm{layer}} = mn + n^2 + n + 11n. \tag{27}$$

Complete network counts include the task readout and use the bias settings of the corresponding layer. A width-n layer stores $5n$ intrinsic state values in addition to its spike cache. Its local update takes $O(n)$ operations per step, while the dense synaptic products take $O(mn + n^2)$.

IV. Experimental Setup

A. Tasks, Encodings, and Data Partitions

S-MNIST presents each MNIST image [27] as 784 scalar inputs, normalized with mean 0.1307 and standard deviation 0.3081. PS-MNIST uses a pixel permutation generated from the run seed and shared between training and test. Variability across seeds on PS-MNIST therefore includes changes in pixel order. Both tasks use two recurrent layers of widths 64 and 210. At evaluation, the linear readout averages logits over the full sequence.

QTDB uses four event streams derived from two ECG channels of the QT Database [28]. The preprocessing code smooths each channel and thresholds its first differences at

± 0.03. The readout predicts six annotation groups at every timestep. Accuracy counts all positions in overlapping windows of length 1301 and stride 650. The generator lists record *sel16483* in both partitions. The record membership of the preprocessed files is unverified, which limits interpretation of this score as generalization to unseen records.

GSC uses 12 classes from Speech Commands [29]. Forty mel coefficients and their first and second differences give 120 features per frame. A 300-unit recurrent layer pools spikes over 101 frames for classification. SHD uses 700 cochlear input channels and 20 digit classes [30]. A 64-unit recurrent layer drives 20 nonrecurrent SPRiF output units. The readout sums the softmax of their pre-reset membrane values over indices $t > 10$.

B. Training and Checkpoint Selection

The benchmark runs use Adam for 150 epochs on S-MNIST, PS-MNIST and GSC, 250 on QTDB and 100 on SHD. The initial learning rate is 0.01 except on GSC, where it is 0.005. Batch sizes are 512 for the MNIST tasks, 128 for QTDB and GSC, and 100 for SHD. On both sequential MNIST tasks, each sequence is split into segments of 262, 262 and 260 steps. Each segment receives an optimizer update. State values carry over, and gradients stop at the boundaries. Within a segment, gradients pass through the spike and reset using Eq. (26). QTDB scores a 31-unit recurrent layer at each timestep. GSC uses negative log likelihood on the pooled readout with class-weighted sampling.

For S-MNIST, PS-MNIST, QTDB and SHD, each run contributes its highest test accuracy across epochs. The reported test split therefore also serves for checkpoint selection. GSC contributes its highest validation accuracy. SI-DMS uses the final checkpoint after 2000 updates.

Each benchmark decay is initialized from a positive time constant in discrete steps:

$$q = \exp(-1/\tau_q), \qquad q_{\mathrm{raw}} = \log \frac{q}{1-q}. \tag{28}$$

Frequency is parameterized analogously through ω/π. All intrinsic states start at zero. The two rows of $\bar{G}$ are optimized directly. The second row uses the product initialization in Eq. (8) on the benchmarks and starts at zero on SI-DMS. On the MNIST tasks the real-mode and filter time constants are sampled uniformly in steps, with the upper bound of the oscillatory time constant decreasing with frequency. QTDB, GSC and SHD use log-uniform time constants. The nominal $(\tau_\alpha, \tau_\rho, \tau_\eta)$ intervals are $([2, 60], [2, 40], [0.8, 10])$ for both MNIST tasks, $([20, 120], [4, 30], [0.8, 8])$ for QTDB, and $([10, 80], [4, 30], [0.8, 8])$ for GSC and SHD. Here τ_η denotes the range for each filter or membrane decay. The respective frequency ranges for ω/π are $[0.04, 0.50]$, $[0.02, 0.20]$ and $[0.04, 0.40]$.

C. Controls with Matched Local Capacity

SPRiF, `MEMORY_RESET5` and `REAL_MODES3` have the same layer widths, five states and eleven learned dynamics coefficients per neuron on PS-MNIST and SHD. We use the same partitions, network readout, training budget, checkpoint rule and seed-specific inputs across variants. The two comparisons alter different terms in the drive to the membrane.

1) `MEMORY_RESET5`*:* Compute the current spectral state $\mathbf{x}_t^-$, both readouts, the auxiliary state, and the membrane spike as in SPRiF. After firing, additionally update

$$\begin{aligned} \widehat{B} &= \frac{B}{\|B\|_2}, \qquad \gamma_{\mathrm{mem}} = 1, \\ \mathbf{x}_t^+ &= \mathbf{x}_t^- - \gamma_{\mathrm{mem}} \theta z_t \widehat{B}. \end{aligned} \tag{29}$$

The next update starts from $\mathbf{x}_t^+$. Because h_t was computed from $\mathbf{x}_t^-$, the extra displacement enters the filter only at the following step. The reset has magnitude θ per spike and introduces no learned coefficient.

2) `REAL_MODES3`*:* Replace the spectral matrices by

$$A_{\mathrm{real}} = \mathrm{diag}(a_1, a_2, a_3), \qquad B_{\mathrm{real}} = \begin{bmatrix} 1 - a_1 \\ 1 - a_2 \\ 1 - a_3 \end{bmatrix}, \tag{30}$$

where $a_i = \sigma(a_{i,\mathrm{raw}})$ are independently learned. Three real decay parameters replace α, ρ, ω. The auxiliary filter, membrane, scalar reset, and six readout entries are retained.

The first real-mode time constant uses the τ_α interval, and the other two use independent draws from the τ_ρ interval. Each coordinate has a nonzero input coefficient. Simply setting the rotation angle to zero would leave one coordinate undriven from a zero initial state.

D. Benchmark Comparators

TC-LIF [8] and BRF [6] were rerun on PS-MNIST and SHD. DMP-SNN [11] was rerun on PS-MNIST using its native architecture with layer-shared memory. Table I distinguishes these reruns from published accuracies and lists total network parameters for each comparison.

E. SI-DMS: Task and Spike Intervention

SI-DMS is a binary match/non-match task. Each trial presents two 100-ms cues separated by a delay, following a 50-ms baseline. Thirty input channels encode the left cue, right cue and background in groups of ten. At each 5-ms step, active cue channels emit a Bernoulli spike with probability 0.20 and background channels with probability 0.05. A recurrent layer of 64 neurons drives two leaky outputs with a 20-ms time constant. The final outputs determine the class.

The training delays are 200, 400, 800 and 1600 ms. Clean training uses natural firing throughout. Intervention training samples $K \in \{0, 1, 2, 4, 8\}$ forced-spike times during the delay. At each selected time, ten of the 64 units (nominally 15%) have their membrane candidates set to the current threshold plus 0.05. The induced spikes trigger each model's reset and enter the recurrent feedback. For SPRiF, the overwrite leaves the current spectral and auxiliary states intact. Training uses Adam with learning rate 0.003, batches of 256 and 2000 updates. Evaluation uses $K \in \{0, 1, 2, 4, 8, 16, 32, 40\}$ and adds a 2500-ms delay. We present the nominal 15% intervention condition, corresponding to ten of 64 neurons (15.625%) at each selected timestep.

TABLE I
CLASSIFICATION ACCURACY AND NETWORK SIZE ON FIVE TEMPORAL BENCHMARKS.

Method	Net	P (K)	Accuracy (%)
S-MNIST			
GLIF [31]	Rec	150	96.64
ASRNN [1]	Rec	150	98.70
DH-SNN [9]	Rec	80	98.90
BRF [6]	Rec	69	99.10
TC-LIF [8]	Rec	150	99.20
DMP-SNN (II) [11]	DMP	73	99.20
SPRiF (ours)	**Rec**	**67**	$\mathbf{99.28 \pm 0.06}$
PS-MNIST			
GLIF [31]	Rec	150	90.47
ASRNN [1]	Rec	150	94.30
DH-SNN [9]	Rec	80	94.52
BRF [6]	Rec	69	95.20
TC-LIF [8]	Rec	150	95.36
DMP-SNN (I) [11]	DMP	61	95.50
TC-LIF^{R} [8]	Rec	150	95.10 ± 0.33
BRF^{R} [6]	Rec	69	95.02 ± 0.41
DMP-SNN^{N} [11]	DMP	67	95.41 ± 0.52
SPRiF (ours)	**Rec**	**67**	$\mathbf{95.86 \pm 0.14}$
QTDB			
ASRNN [1]	Rec	1.8	85.90
DH-SNN [9]	Rec	1.78	86.35
BRF [6]	Rec	1.73	87.00
SE-adLIF [2]	Rec	1.8	86.88
SPRiF (ours)	**Rec**	**1.7**	$\mathbf{88.43 \pm 0.36}$

Method	Net	P (K)	Accuracy (%)
GSC			
SNN-SFA [32]	Rec	4190	91.21
ASRNN [1]	Rec	310	92.10
RadLIF [33]	Rec	1200	94.51
DH-SNN [9]	Rec	130	93.80
TC-LIF [8]	Rec	200	94.84
SPRiF (ours)	**Rec**	**130**	$\mathbf{94.55 \pm 0.31}$
SHD			
Heterogeneous SNN [3]	Rec	110	82.50
ASRNN [1]	Rec	140	90.40
DH-SNN [9]	Rec	50	91.34
BRF [6]	Rec	110	91.70
TC-LIF [8]	Rec	140	88.91
MPS-SNN [34]	Rec	390	91.19
DGN [35]	Rec	220	87.78
TC-LIF^{R} [8]	Rec	140	88.73 ± 0.44
BRF^{R} [6]	Rec	110	91.28 ± 0.53
SPRiF (ours)	**Rec**	**50**	$\mathbf{91.66 \pm 0.42}$

Notes: P (K): parameters in thousands; Rec: recurrent network; DMP: dual-memory-pathway architecture. Unmarked baselines are published results; ^{R}our reruns; ^{N}our full DMP-SNN run. Our results are mean $\pm$ sample SD over five seeds. GSC uses best validation accuracy; other tasks use best test accuracy across epochs.

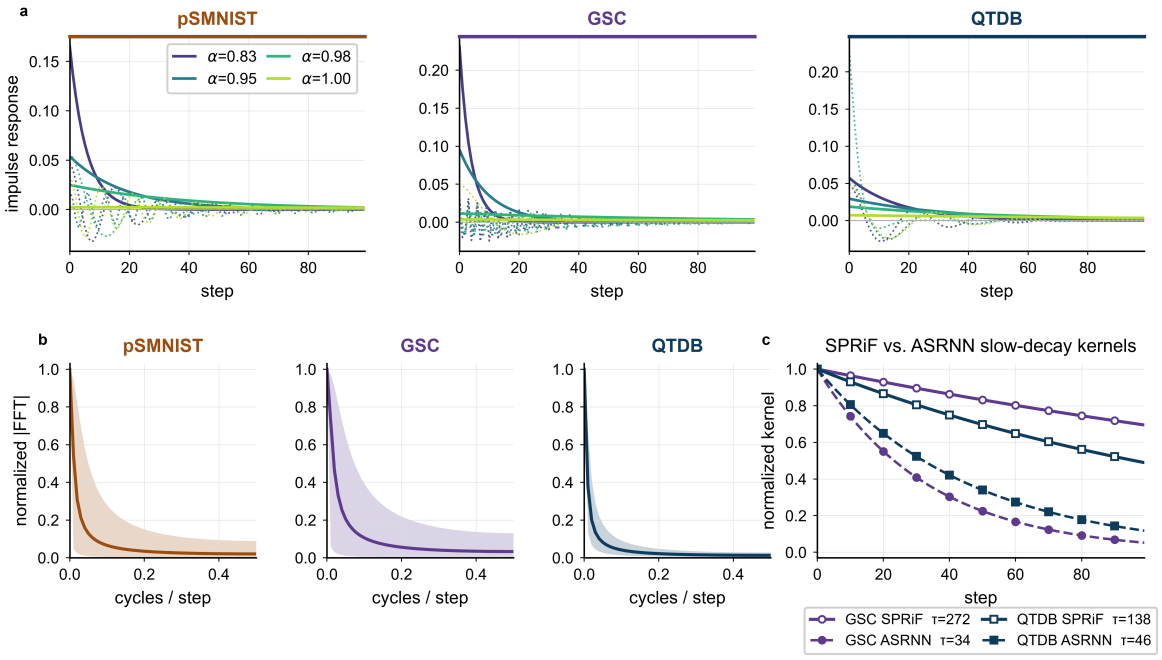


Fig. 2. Temporal responses of SPRiF across tasks, arranged in three parts with seven subplots. (a) The three upper subplots show SPRiF impulse responses on PS-MNIST, GSC and QTDB, from left to right. (b) The three lower-left subplots show normalized frequency-response magnitudes of the memory states for the same tasks. Shading indicates the distribution range across neurons, not uncertainty across random seeds. Frequency is expressed in cycles per step. (c) The lower-right subplot compares normalized slow-decay responses of SPRiF and ASRNN on GSC and QTDB. Solid curves with open markers denote SPRiF; dashed curves with filled markers denote ASRNN.

For seed s, delay d, count K, and fraction f, the paired loss is

$$\Delta_{s,d}(K, f) = \mathrm{Acc}_{s,d}(0) - \mathrm{Acc}_{s,d}(K, f). \tag{31}$$

At $K = 40$ and nominal $f = 0.15$, we report the loss in percentage points over all five evaluation delays and separately at 2500 ms.

F. Statistical Reporting

One training run is one replicate. We report mean accuracy and sample standard deviation across seeds $\{0, 1, 2, 42, 1111\}$. Published entries retain their original reporting conventions.

TABLE II
STRUCTURAL ABLATIONS. THE THREE VARIANTS USE THE SAME LAYER WIDTHS, FIVE STATES AND ELEVEN LEARNED LOCAL COEFFICIENTS PER NEURON. ACCURACY IS MEAN ± SAMPLE SD (%) OVER FIVE SEEDS. THE SPRIF REFERENCE REUSES TABLE I.

Variant	PS-MNIST	SHD
SPRiF	95.86 ± 0.14	91.66 ± 0.42
`MEMORY_RESET5`	92.49 ± 0.36	87.10 ± 0.61
`REAL_MODES3`	93.52 ± 0.27	89.19 ± 0.50

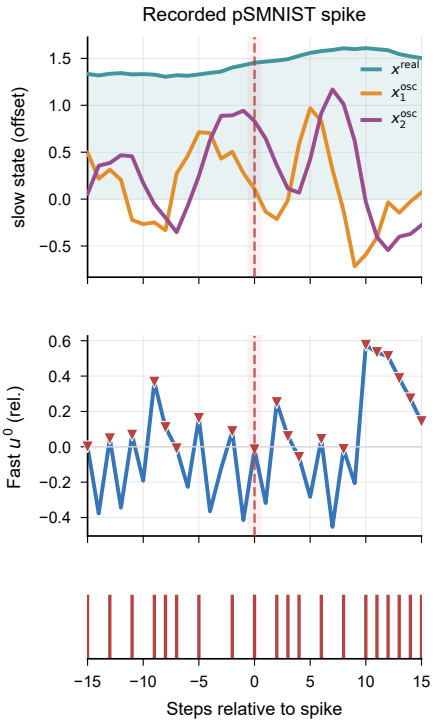


Fig. 3. SPRiF states before and after a recorded spike on PS-MNIST. The upper panel shows the real and two oscillatory memory coordinates, vertically offset for visibility. The middle panel shows relative membrane voltage, and the lower panel shows the spike train. The vertical dashed line marks the reference spike at relative step zero. The aligned traces show how firing and membrane reset relate to the continuing evolution of the internal memory states.

In SI-DMS, each delay has equal weight within a seed. We report the mean Δ_{40}, equal to mean clean accuracy minus mean stressed accuracy.

V. RESULTS AND DISCUSSION

A. Temporal Classification at the Reported Network Sizes

Table I summarizes five temporal benchmarks. On PS-MNIST, SPRiF reached $95.86 \pm 0.14\%$ with 67,000 parameters, compared with $95.41 \pm 0.52\%$ for the native DMP-SNN rerun at the same reported size. On SHD, SPRiF reached $91.66 \pm 0.42\%$ with 50,000 parameters. The BRF and TC-LIF reruns reached $91.28 \pm 0.53\%$ and $88.73 \pm 0.44\%$ with 110,000 and 140,000 parameters, respectively. The controls below test how SPRiF uses its states on these two tasks while keeping layer width and local capacity fixed.

B. Temporal Responses of the Spectral Memory

Figure 2 connects the spectral basis in Eq. (10) to the temporal responses on PS-MNIST, GSC and QTDB. The impulse responses show a real decay alongside damped oscillatory components (Fig. 2a). These components provide traces with different durations and phases for the membrane projections in Eq. (13). The frequency responses summarize the memory states across neurons in each task (Fig. 2b). Their shaded ranges describe variation across neurons. In the slow-decay comparison, the displayed SPRiF responses retain a larger normalized amplitude than ASRNN over the plotted interval on GSC and QTDB (Fig. 2c).

C. Reset Placement and the Choice of Modes

Figure 3 shows SPRiF memory states, membrane voltage and spikes around a recorded firing event on PS-MNIST. The memory trajectories continue through the marked event, while the membrane trace reflects firing and reset. This illustrates the separation in Eq. (4): the spike changes the stored membrane voltage without subtracting from the spectral memory.

The two neurons have the same scalar membrane reset. In `MEMORY_RESET5`, a spike at t also contributes $r[1] = -\theta \mathbf{g}_0^\top A\widehat{B}$ to the drive at $t+1$ (Eq. (23)). The filter first carries part of that displacement to the membrane at $t+2$. Its sign depends on the learned projections. A memory reset thus adds a spike-dependent signal to the readout even if the incoming current is held fixed. SPRiF omits this path and scored 3.37 percentage points higher on PS-MNIST and 4.56 points higher on SHD (Table II).

The mode comparison keeps the reset on the membrane. In `REAL_MODES3`, the readout combines three driven exponential decays. In SPRiF, an input also leaves a trace whose direction rotates with elapsed time. Its projection can therefore weight successive phases differently under a common decay envelope (Eq. (10)). The real-plus-rotational block scored 2.34 percentage points higher on PS-MNIST and 2.47 higher on SHD than the three-real-mode block. This comparison tests the choice of basis and its associated input coupling together.

D. Forced Spikes During a Memory Delay

SPRiF maintained high SI-DMS accuracy as forced spikes accumulated during the memory delay (Fig. 4). After clean training, its five-delay mean accuracy decreased from 99.99% at $K = 0$ to 99.42% at $K = 40$. The corresponding stressed accuracies were 94.74% for `MEMORY_RESET5` and 95.37% for `REAL_MODES3`, compared with 48.56–54.80% for ASRNN, BRF and LIF. SPRiF therefore combined accurate cue retention with the highest mean accuracy under strong intervention (Table III).

Intervention-trained SPRiF retained $99.33 \pm 0.78\%$ accuracy at $K = 40$, with a mean decrease of 0.36 percentage points from $K = 0$. Its high accuracy under both training regimes shows that tolerance to these interventions did not require exposure during training. The two matched controls reached lower stressed means and exhibited greater variation across seeds after intervention training. In particular, `REAL_MODES3`

TABLE III

SI-DMS UNDER CLEAN AND INTERVENTION TRAINING. STRESS USES $K = 40$ AND TEN OF 64 NEURONS PER INTERVENTION (NOMINALLY 15%). ACCURACY IS MEAN ± SAMPLE SD (%) ACROSS FIVE TRAINING SEEDS. Δ_{40} IS THE MEAN CLEAN-MINUS-STRESS DIFFERENCE IN PERCENTAGE POINTS. THE FIVE-DELAY SUMMARY INCLUDES THE UNSEEN 2500-MS DELAY, ALSO SHOWN SEPARATELY.

		Five-delay average			2500 ms		
Training	Model	$K = 0$	$K = 40$	Δ_{40}	$K = 0$	$K = 40$	Δ_{40}
Clean	SPRiF	99.99 ± 0.01	99.42 ± 0.69	0.57	99.99 ± 0.01	99.99 ± 0.01	0.00
	MEMORY_RESET5	99.93 ± 0.07	94.74 ± 2.48	5.19	99.93 ± 0.10	99.84 ± 0.13	0.09
	REAL_MODES3	99.90 ± 0.20	95.37 ± 5.96	4.53	99.91 ± 0.16	99.87 ± 0.16	0.04
	ASRNN	99.94 ± 0.03	48.56 ± 1.99	51.39	99.89 ± 0.13	47.50 ± 4.48	52.39
	BRF	82.00 ± 0.76	54.80 ± 1.65	27.19	61.87 ± 4.95	50.62 ± 0.97	11.25
	LIF	88.46 ± 18.87	51.01 ± 1.56	37.44	83.35 ± 20.95	53.36 ± 7.55	30.00
Intervention	SPRiF	99.69 ± 0.67	99.33 ± 0.78	0.36	99.65 ± 0.71	99.65 ± 0.72	0.00
	MEMORY_RESET5	94.46 ± 12.23	88.54 ± 20.25	5.92	93.81 ± 13.75	91.23 ± 19.32	2.57
	REAL_MODES3	89.94 ± 22.47	89.91 ± 22.45	0.03	90.08 ± 22.16	90.04 ± 22.13	0.04
	ASRNN	53.22 ± 3.17	49.85 ± 0.13	3.37	49.94 ± 0.26	49.80 ± 0.77	0.13
	BRF	73.28 ± 3.42	51.34 ± 0.48	21.94	57.15 ± 7.99	50.11 ± 0.40	7.04
	LIF	56.53 ± 1.42	49.98 ± 0.24	6.55	50.09 ± 0.61	50.07 ± 0.83	0.02

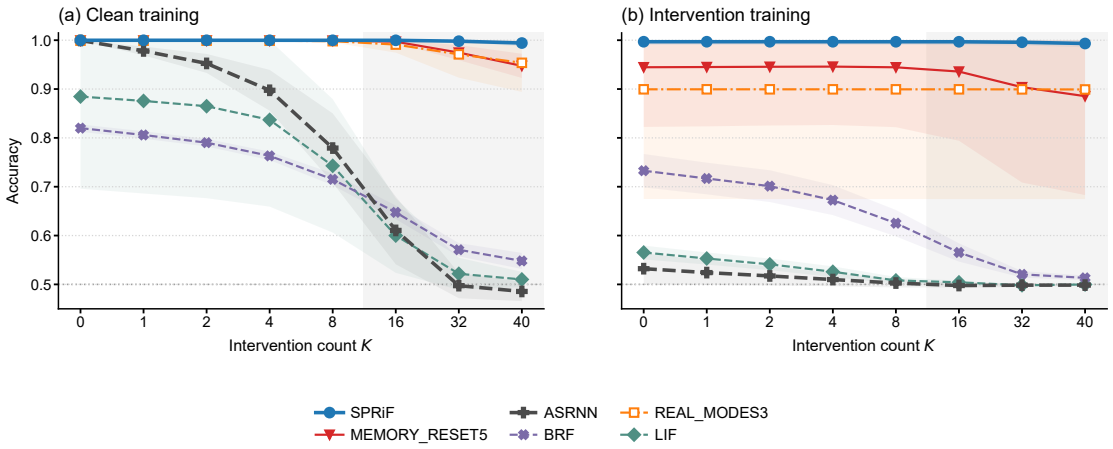


Fig. 4. SI-DMS accuracy as the number of forced-spike timesteps increases, following (a) clean training and (b) intervention training. Each intervention targets ten of 64 neurons. Curves show means over five training seeds; shading indicates ± one sample SD, clipped to the accuracy range. Within each seed, the five evaluation delays receive equal weight. The dotted line denotes 50% chance accuracy. The gray region marks intervention counts beyond the maximum used in intervention training ($K = 8$). Horizontal positions are equally spaced evaluation conditions.

had a nearly flat mean curve but a sample SD of approximately 22.45 percentage points at $K = 40$. A small within-model decrease therefore needs to be read alongside absolute accuracy and variation across runs.

Figure 5 resolves the delay dependence hidden by the averaged curves. SPRiF maintained seed-averaged accuracy above 97.7% at every displayed delay–intervention combination under both training regimes. At the unseen 2500-ms delay and $K = 40$, it reached 99.99% after clean training and 99.65% after intervention training. The heatmaps also show that longer delays were not uniformly harder: the clean-trained matched controls lost more accuracy at short delays under strong intervention.

The paired replay in Fig. 6 illustrates how activity and decisions respond to an intervention within one SPRiF network. For the same A–A input, forcing spikes at eight delay timesteps changed the spike pattern, while the final readout still favored a match. This behavior occurred in both displayed checkpoints. The second-cue activity traces differed transiently, but the match probabilities approached the same endpoint. These paired trajectories connect the aggregate accuracies to the network's response on an individual trial.

The local equations explain why preserving spectral memory can support this behavior without making network activity invariant to intervention. A forced discharge in SPRiF leaves the current spectral and auxiliary coordinates intact, while changing the membrane and potentially the recurrent input on the next step. The additional direct memory-reset path in MEMORY_RESET5 remains absent (Eq. (24)). Thus the observed robustness concerns cue comparison despite altered spike histories, rather than identical trajectories.

VI. CONCLUSION

SPRiF gives the membrane a decaying trace and a phase-dependent trace of past input, each available through an immediate and a delayed projection. Each spike resets the membrane and can re-enter the spectral pathway one step later through recurrent current. Extending reset to spectral memory adds a distinct spike-history term to the drive. At fixed local capacity, both preserving that memory at discharge and retaining its

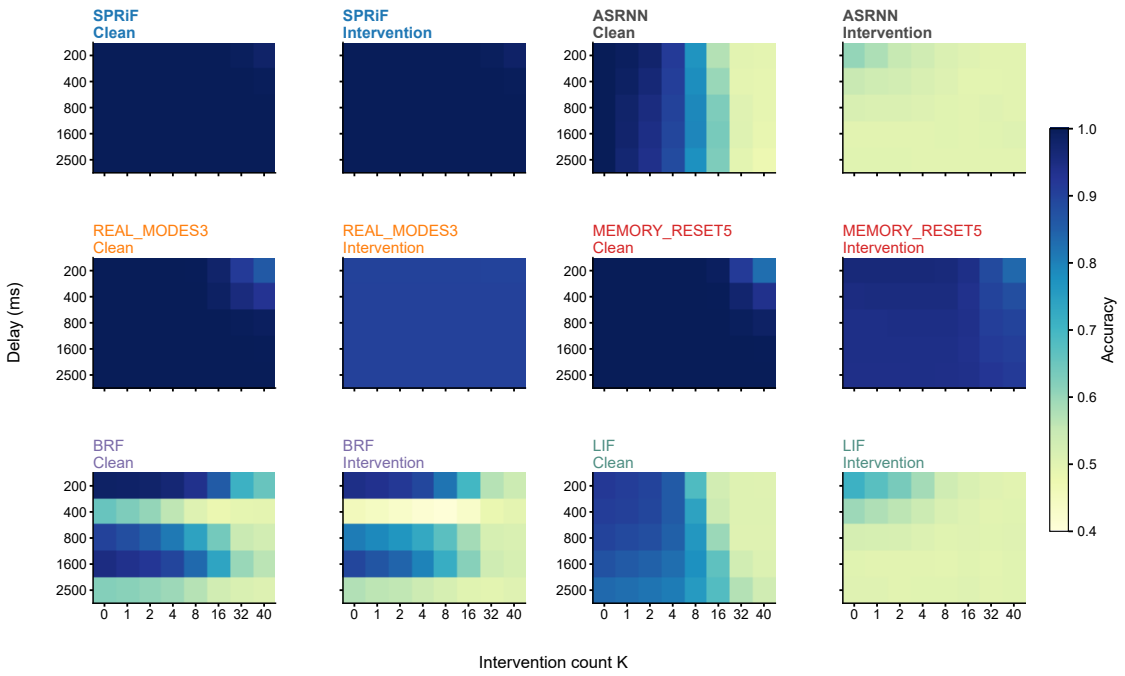


Fig. 5. SI-DMS accuracy across delays and intervention counts, with adjacent Clean and Intervention training panels for each model. Cells average five seeds; the shared scale spans 40%–100%. Each intervention targets ten of 64 neurons; $K = 0$ is unforced. The 2500-ms delay and counts $K = 16, 32, 40$ lie outside the intervention-training ranges.

rotational pair improved mean accuracy on PS-MNIST and SHD. On delayed match-to-sample, SPRiF sustained accurate cue comparison under both clean and intervention training, including an unseen delay and intervention counts beyond the training range.

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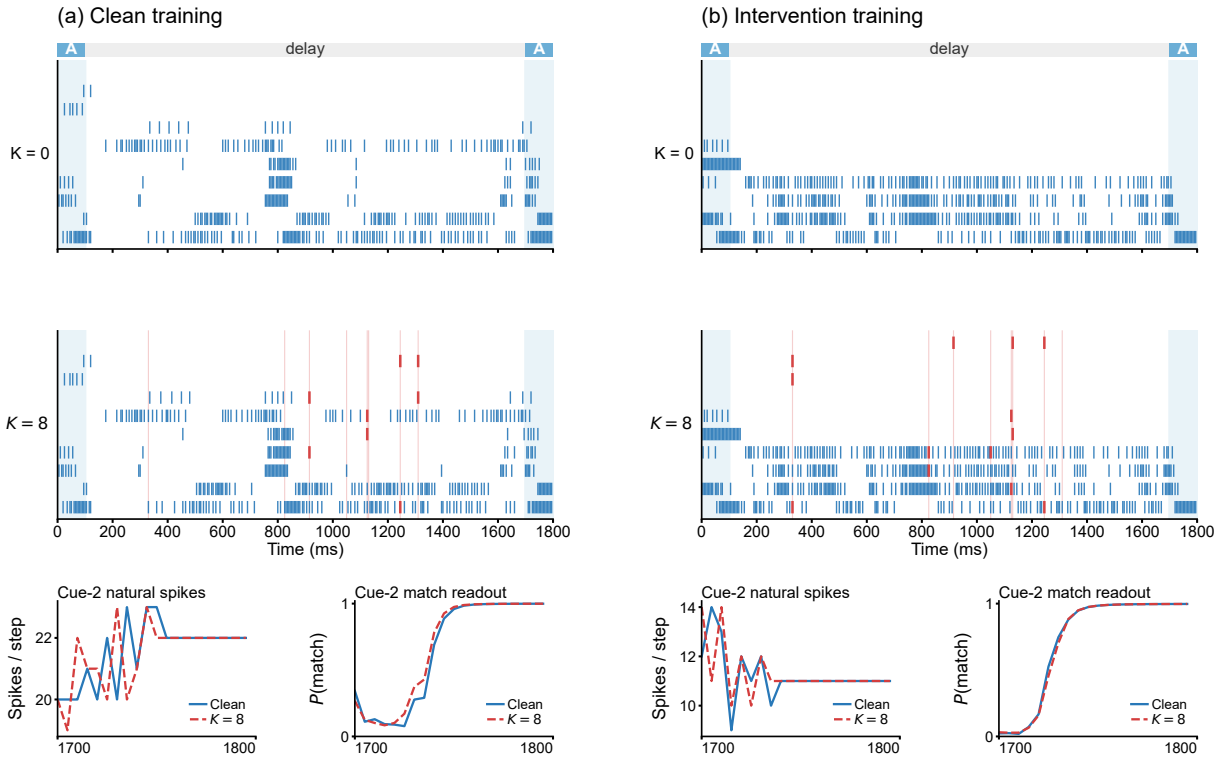


Fig. 6. Paired SPRiF replays from seed-0 checkpoints after (a) clean and (b) intervention training. Both use the same A–A input and 1600-ms delay. Rasters compare $K = 0$ and $K = 8$ for ten neurons, fixed within each checkpoint. Blue/red ticks denote natural/forced spikes; pale blue bands mark cues. Bottom panels show second-cue natural spike counts across all 64 neurons and match probabilities: solid blue, $K = 0$; dashed red, $K = 8$. Time starts at the first cue.

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